Decision Tree 2

Saturday, February 24, 2024 9:01 PM

DT → 1

- Rules – if else
- Impurity → Best or Top Root question

Purity

① Entropy → Information Gain

② overfitting

Test 70%

Training 100%

③ Imbalance Data

④ Scaling

⑤ Feature Imp ★ ★ ★

⑥ DT → Regression – 3

Entropy

$$-\sum p(y) \log_2(p(y))$$ DT

$$-\sum p(y)\log_2(p(y))$$ DT

1. entropy for all the features
2. IG
3. Max IG Root Node

log → 1 DT RF → 100 DT

↓ High computation

Gini Impurity

$$1-\sum_{i=1}^{k} p(y)\times p(y)$$

$$1-\sum_{i=1}^{k} p(y)^2$$

} Gini Impurity

Comparing Gini Impurity & Entropy

* Remember the 6 Red & blue ball example

Red Blue

Entropy

Entropy

Entrop

$\frac{1}{2}\log\frac{1}{2} +$

$P(y)$

$\left(\frac{3}{6}\right)^{\times}$

* Remember the 6 Red & blue ball example

Red Blue

C1: $GI(y) = 1 - \left[\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right] = 0.5$; $H(Y) = \left[\left(\frac{1}{2}\right)\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right] = 1$

C2: $GI(y) = 1 - \left[\left(\frac{1}{6}\right)^2 + \left(\frac{5}{6}\right)^2\right] = 0.27$; $H(Y) = \left[\frac{1}{6}\log_2\frac{1}{6} + \frac{5}{6}\log_2\frac{5}{6}\right] = 0.65$

C3: $GI(y) = 1 - \left[\left(\frac{0}{6}\right)^2 + \left(\frac{6}{6}\right)^2\right] = 0$; $H(Y) = [0\log_2 0 + 1\log_2 1] = 0$

2 class

Highest Gini = 0.5 } Entropy = 1

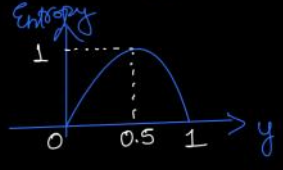
Lowest Gini = 0 } entropy = 0

Gini Impurity follows same trend as Entropy

* Meaning ↑ Impurity → ↑ Gini Impurity

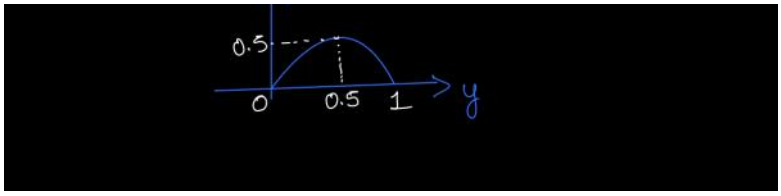
Major difference between Entropy & Gini Impurity

* Range of Entropy [0, 1]



* Range of Gini Impurity [0, 0.5]





IG = Parent Entropy − Weighted Child Entrop

IG = Parent Gini Idex − weighted Gini Index child

Class A | Class B

60 | 40

$$1 - \left[\left(\frac{60}{100}\right)^2 + \left(\frac{40}{100}\right)^2 \right]$$

$$1 - [0.36 + 0.16]$$

$$1 - 0.52$$

$$\boxed{0.48}$$

How to work categorical data

Gender = Male

yes — No

How to work on Numerical Data

$age \leq 35$ $\quad T < 22°C$

f_1 — Temp } Threshold

f_2 — Windspeed

f_3 → Wheather — Rainy, sunny, cold — IG

Step1

Temp

n_1
n_2
n_3
n_4
.
.
.
n_m

unique Value in ascending order

$D_1 \rightarrow$ Temp $\leq n_1$ → Gini Impurity

yes ← 60 , → No 40 → 0.48 } IG

→ Temp $\leq n_2$ → Gini I } IG

Highest IG

35
75

Problem with threshold approach

if there are 3 numerical features each with 1000 unique values:

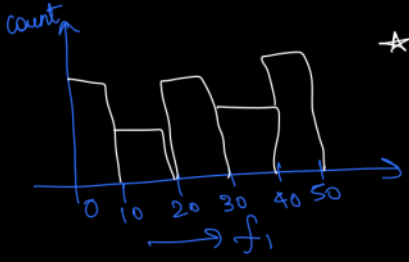
time complexity ≃ O(no of feature × unique value)

which will be huge

How to make splitting of numerical feature efficient?

Using bins for range of feature value:

* Suppose f_1 value range: [0, 50]



* We create 5 bins for f_1 value & check IG value for these bins
* Select the bin for splitting which has maximum IG

Tab

25% 33 IG

5%
10%

3
10
12

25 %. 33 } IG

50 %. 22 } IG

75 %. 11 } IG

100 %. 3 } IG

10 %.

15 %.

20 %.

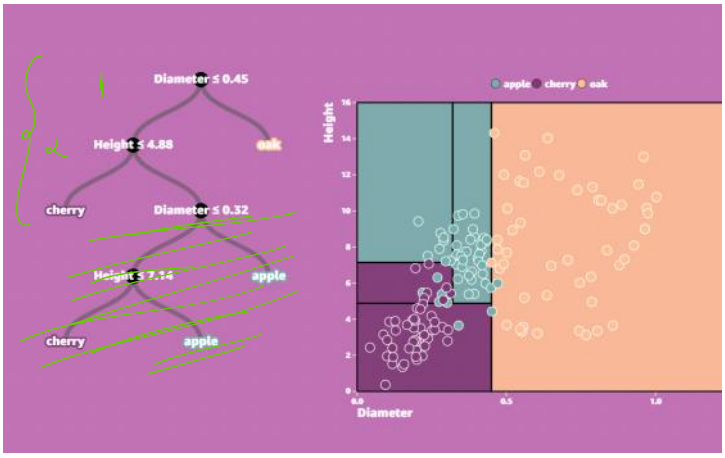
25 %.

10
12
22
21
~~27~~

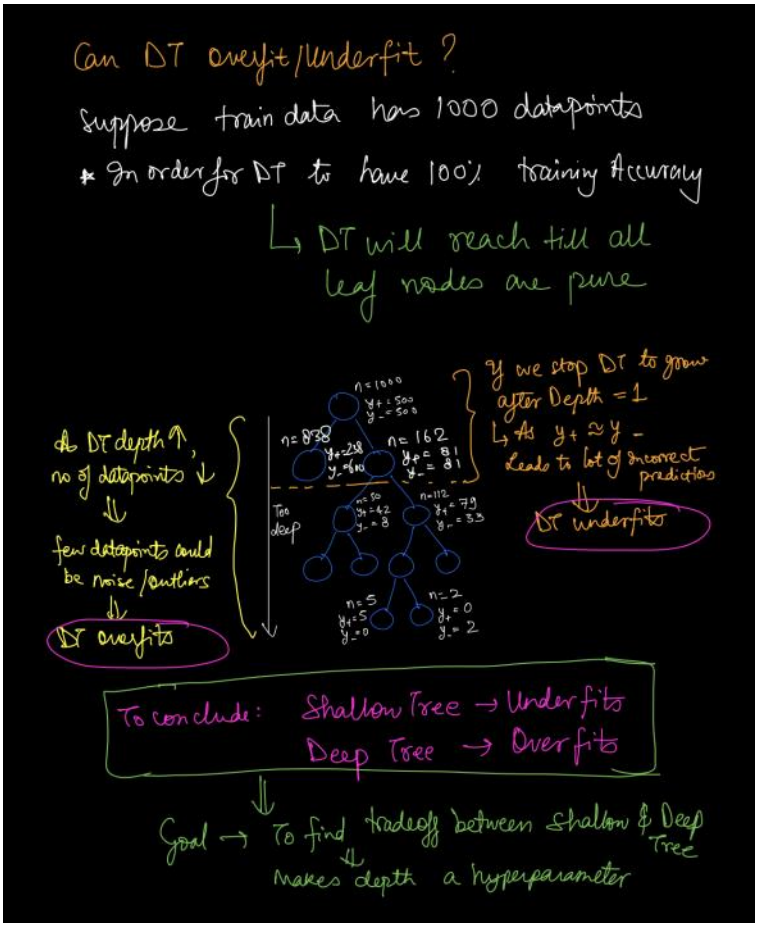
Overfitting

Tree grow

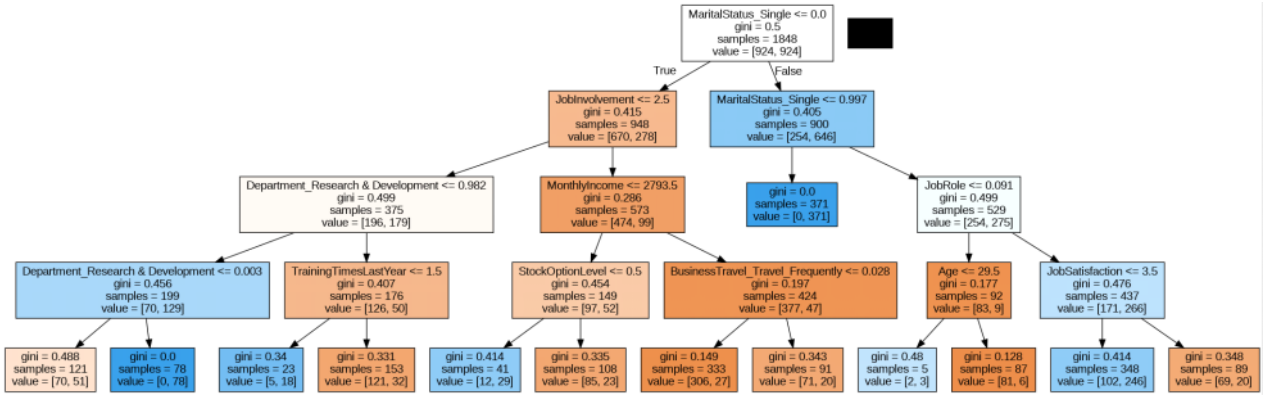
Depth 2

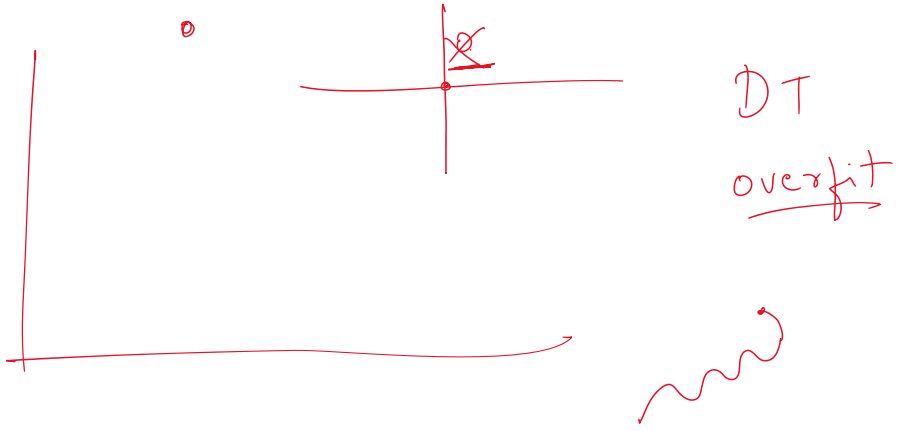


Pruning of Tree



{Case study} {Practice} { Logistic - Imbalance dataset { 0:1 1:5.72 } }





ML { feature Scaling
all features should be in same range

KNN SVM LR

New Section 20 Page 9

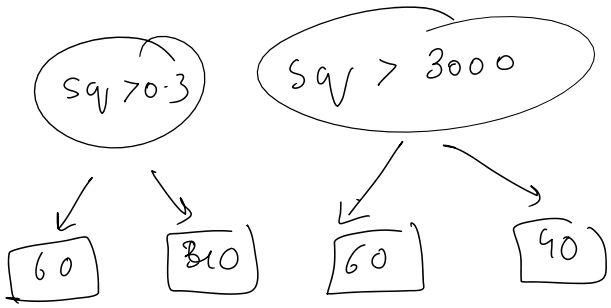
DT —— ?

sqm

1000 → 0.1

3000 → 0.3

7000 → 0.7



Does DT require feature scaling?

	f_i		
n_1	2.1	⟹ Scaled to	-1
n_2	2.6		-1.2
n_3	2.8		0

Suppose we scale feature f_i

* As impurity measure depends on no of samples of classes & not the value
* Hence, even if we scale the feature f_i, does not effect DT, hence feature scaling not required.

DT works well on Categorical feature without any Encoding

Dataset — Ensemble }

Encoding of categorical variable

Encoding of categorical

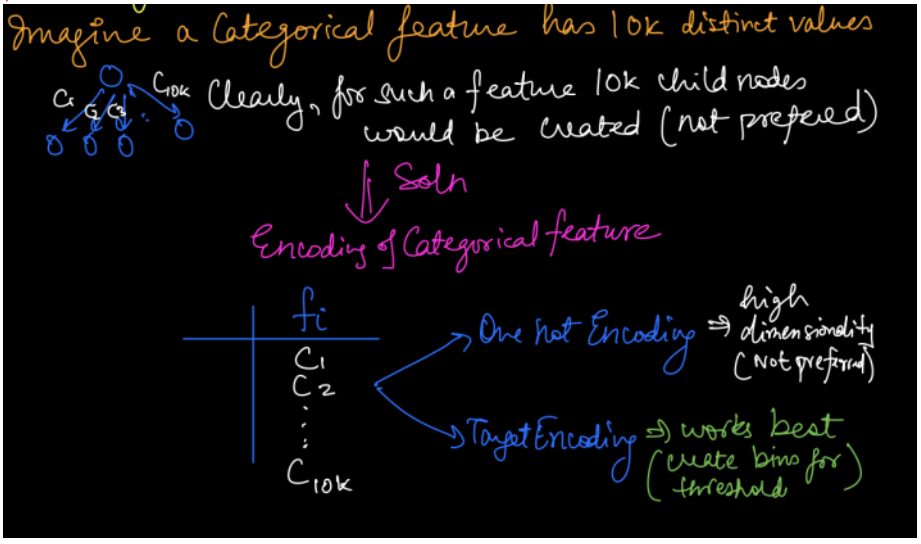
yes

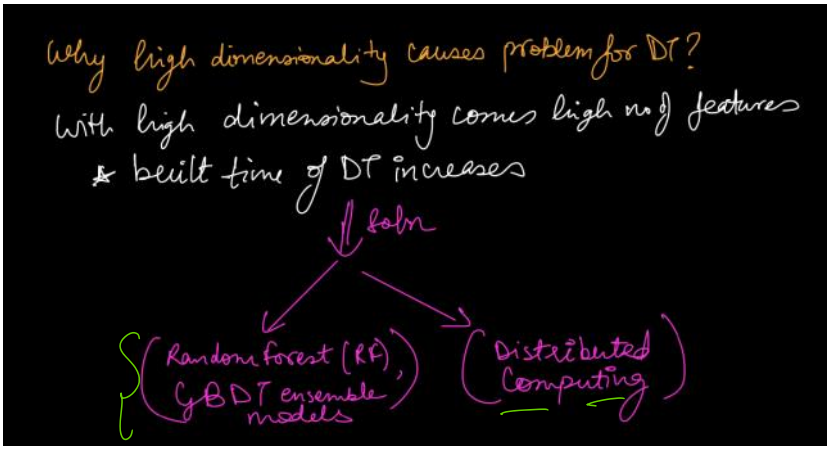
OHE

MF

Target Encoding.

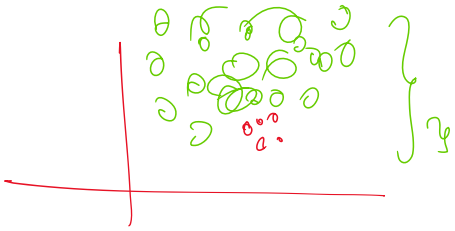
1000 unqu





Imbalance Data → Decision Tree.

↓ depends on number of Data point per class



Does imbalance in data impacts DT?

DT goal is to minimize impurity → depends on the number of data points per class

In imbalance setting
* DT aims to correctly classify majority class
* At the cost of poor performance for minority class

↓ Soln

use data balancing techniques
- → class weights
- → Smote
- → Undersampling / Oversampling

Feature Imp

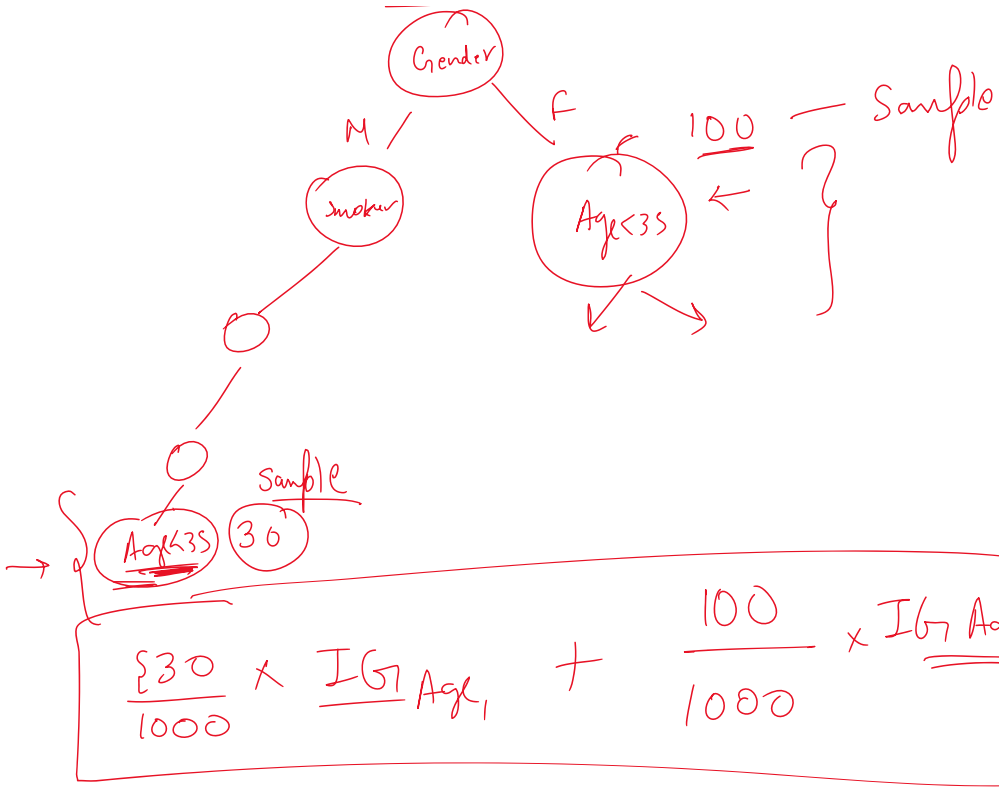
36 → column

- → 1
- → 2
- 3

Leave Perk

} JIO ML

→ 1000



Gender → M — 0
F — 1

unqi

31 − un — ~~OHE~~ — Target

30 col

DT regression

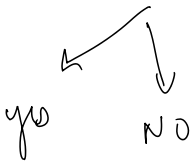
f1	f2	....	Label
			3.2
			7.1
			8.9

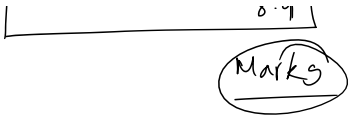
Marks

DT

Target var is continous

f1 < T





DT

continous

DT

- impurity
- Ques → split the node → Reducing Impurity
- Make prediction

Calc Impurity for continous data

How does DT used in Regression?

Remember in regression setting

* target variable (y) becomes continous

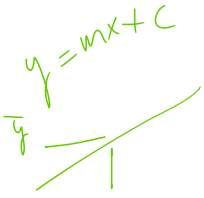
for Regression, will entropy / Gini impurity work? → X as the predictions are not probabilities

↓ Soln

node with samples n_1 Using $MSE_{node} = \frac{1}{n_1} \sum_{i=1}^{n_1} \left(y^{(i)} - \hat{y}^{(i)} \right)^2$

MSE

$$\frac{1}{n} \sum \left(y - \hat{y} \right)^2$$

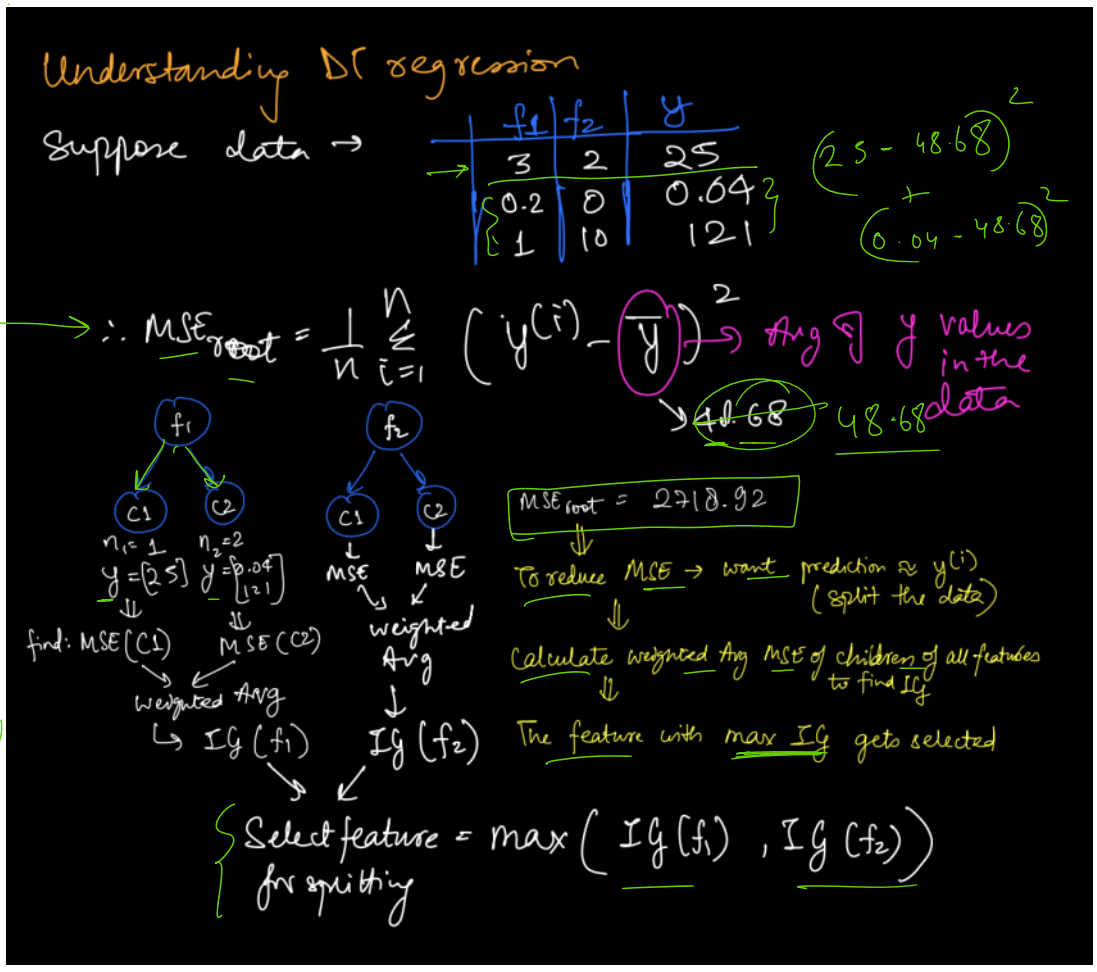


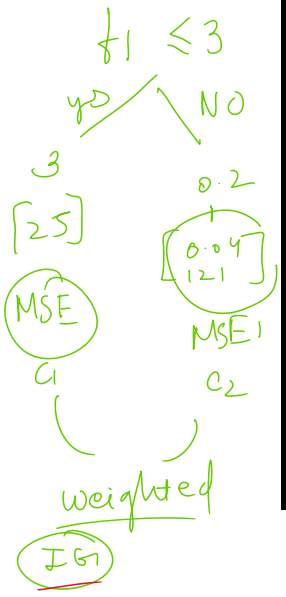
Regression does not have prob

f1	f2	f3	Label
			72
			71
			35
			45

Mean $\bar{y}_{mean}$ 45

$$\frac{\sum (72-45)^2}{n}$$







Definety

{Material, LSH}

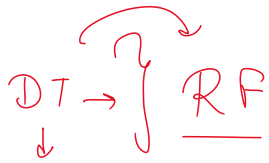
Port Read

Tem 3-4

weighted KNN

3

In



Case study →